Chapter 04

Virtualization & Backup

Episode 4.01

Episode title: **Virtualization Concepts**

Objective: 1.7

Virtualization

Full virtualization

Para-virtualization

Hardware-assisted virtualization

Bare-Metal vs Virtualized Systems

Bare-metal servers

Hypervisors

Virtual machines

Resource utilization

Why Virtualization Is Used

Efficiency

Isolation

Flexibility

Scalability

Cost control

Types of Virtualization

Full virtualization

Para-virtualization

Hardware-assisted virtualization

Linux Virtualization Content & Exam Focus

Linux as host

Linux as guest

Cloud platforms

Exam expectations

Nested Virtualization

Running virtual machines inside VMs

Requires hardware virtualization support

Common in labs and cloud platforms

Exam-level conceptual topic

Episode 4.02

Episode title: **Virtualization Concepts**

Objective: 1.7

Linux Hypervisors

KVM

QEMU

libvirt

KVM and QEMU Roles

KVM kernel modules

QEMU userspace

Hardware acceleration

VirtIO and Paravirtualized Drivers

- VirtIO provides paravirtualized drivers
- Improves VM performance
- Used for disk, network, and memory
- Common with KVM/QEMU

libvrt Management Layer

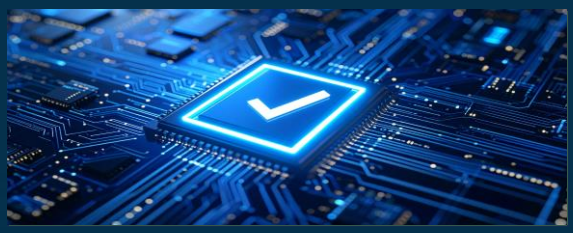
- libvirtd service
- virsh
- virt-install
- virt-manager

Installing and Verifying Linux Virtualization

CPU flags

Kernel modules

Required packages



Linux Hypervisors with KVM, QEMU, and libvirt

Prerequisites

Exact commands

Verification

Cleanup

Episode 4.03

Episode title: **Virtual Machine Operations**

Objective: 1.7

Introduction

Operations

- States
- Resources
- Snapshots
- Migrations

VM Lifecycle & States

Running, paused,
shut off, crashed

start, shutdown, destroy, reboot

Safe transitions

Resource Management & Right-Sizing

vCPU, memory, ballooning disk allocation, thin vs thick networking & NIC types

Virtual Machine Network Types

- Bridged networking
- Network Address Translation (NAT)
- Host-only or isolated networks
- Routed networking
- Open network

Snapshots & Backups

Difference between snapshots and backups

Consistent snapshots

Snapshot management with

virsh/qemu-img

Disk Image Operations with qemu-img

qemu-img manages VM disk images

View image properties

Convert between formats

Resize virtual disks

VM Migration Concepts

Live migration

Cold migration

Shared storage vs block copy

Network and compatibility considerations

Baseline Images, Templates, and Cloning

- Baseline or golden images
- Templates for rapid deployment
- VM cloning concepts
- Consistency and efficiency benefits

VM Operations in Linux (DEMO)

Prerequisites

Exact commands

Verification

Cleanup

Troubleshooting

Episode 4.04

Episode title: **Backup Strategies**

Objective: 1.6

Backup Objectives and Requirements

Data protection

Recovery Time Objective (RTO)

Recovery Point Objective (RPO)

Common Backup Models

Full backups

Incremental backups

Differential backups

Backup Scope and Considerations

What to back up

On-site vs off-site storage

Retention policies

Backup Testing and Recovery Planning

Test restores

Disaster recovery planning

Documentation

Restore Operations – Exam Focus

- File-level restores vs full system restores
- Restore validation and verification
- Matching restore method to backup type
- Exam focus: choosing the correct tool

Episode 4.05

Episode title: **Backup Tools**

Objective: 1.6

Common Tools

- rsync
- tar
- dd

rsync for File-Based Backups

Incremental transfers

Local and remote sync

Preserving permissions

tar and Compression Utilities

Archiving files

Compression formats

Portability

Archiving with cpio

`cpio` – Archive files from standard input

Often used with find

Alternative to `tar`

Exam-level conceptual knowledge

Compression Tools – Exam Coverage

- `gzip` – Fast, common compression
- `bzip2` – Better compression, slower
- `xz` – High compression ratio
- `unzip` – Extract ZIP archives
- `7-Zip` – Cross-platform archives

dd and Raw Disk Backups

Block-level copying

Disk imaging

High risk, high power

Advanced Backup & Recovery Tools

- `ddrescue` – Recover data from failing disks
- `zcat` – View compressed files
- `zgrep` – Search compressed files
- `zless` – Page through compressed files

Backup Tools (rsync, tar, dd) (DEMO)

Prerequisites

Exact commands

Verification

Cleanup